

节能电扇

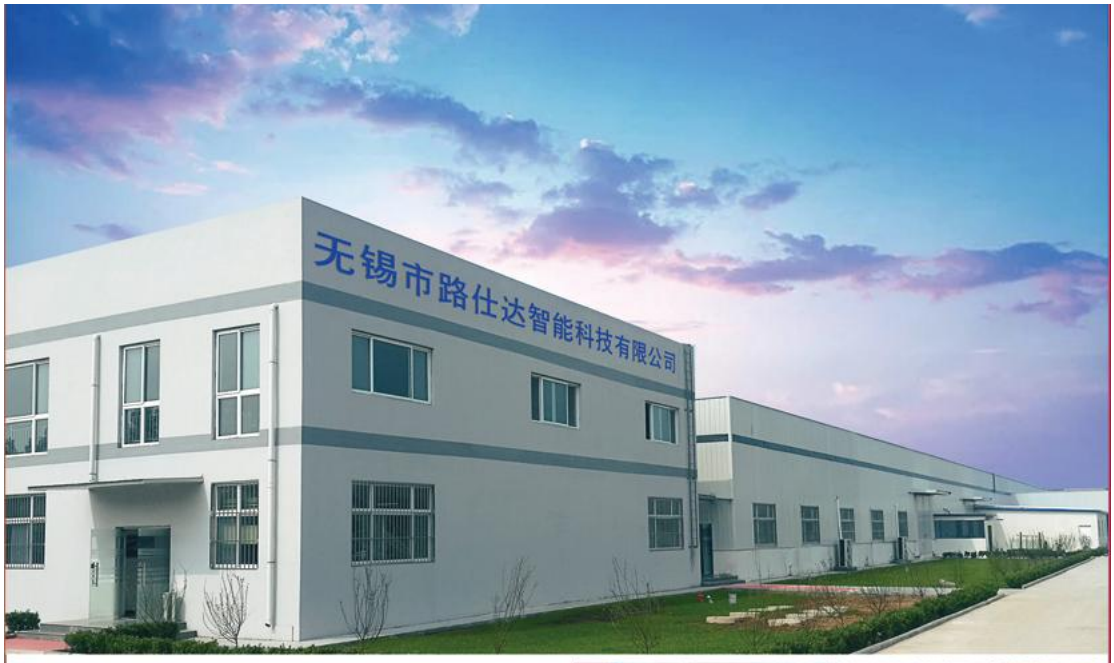
PRODUCT MANUAL

通风/降温解决方案

VENTILATION COOLING SOLUTION

无锡路士达智能科技有限公司

Wuxi Lusida Intelligent Technology Co., Ltd



企业简介

ENTERPRISE INTRODUCTION







无锡市路士达智能科技有限公司位于江苏省无锡市锡山区东港镇里国工业园内，地处长三角经济圈，仅靠沿江高速、沪宁高速、无锡机场、无锡高铁站、火车站、交通十分便利。

公司成立于2019年，主导产品是电机无刷控制器系列的应用，范围涉及无刷风机系列产品。

公司最新研发的一种基于PMSM无刷直驱技术的新型工业吊扇;其最大直径可达7.3米。该产品结合了空气动力学、传动力学、脉宽调制控制技术、机械力学、仿真模拟技术、通讯控制、工业设计等一系列综合学科的前沿技术， 采用国内外先进的精密加工设备制造而成。其可在更高效率下促使空间气流流动循环,达到人员降温的目的，改善环境舒适度。

Wuxi Lushida Intelligent Technology Co. , Ltd. is located in Donggang Liguo Industrial Park, Xishan District, Wuxi City, Jiangsu Province, only rely on the river, Shanghai-nanjing high-speed, Wuxi Airport, Wuxi high-speed railway station, railway station, traffic is very convenient.

The company was founded in 2019, the leading products are brushless motor controller series of applications, involving the range of brushless fan series products. The company's latest research and development based on PMSM brushless direct-drive technology of a new industrial ceiling fan; its maximum diameter of 7.3 meters. The product combines the cutting-edge technologies of aerodynamics, transmission dynamics, pulse width modulation control technology, mechanical mechanics, simulation technology, communication control, industrial design, etc. , it is made of advanced precision machining equipment at home and abroad. The utility model can promote the air flow circulation in the space under higher efficiency, achieve the goal of personnel cooling, and improve the environmental comfort.

我们用实力说话 多元化布局

WE SPEAK WITH STRENGTH, DIVERSIFIED LAYOUT

产品简介
PRODUCT PROFILE



大型工业吊扇最早于1991年诞生于美国，当时的主要作用是农场里面专门用来给牲畜通风的。高温的夏季使农场闷热，臭味，飞虫增多，从而导致农场产量下降,农场主效益不佳。闲话中说起要是放一架直升机就所有问题都解决了,农场主付诸行动,成为了现在大型工业吊扇的原型。

在使用过程中，大型工业吊扇的优势开始显现出来，其在降温,舒适度,通风效果,覆盖范围以及节能方面都比普通小风扇优秀得多。

Large industrial ceiling fans first appeared in the United States in 1991, when they were used to ventilate livestock on farms. The high temperature summer makes the farm muggy, the odor, the flying insect increase, thus causes the farm output to drop, the farmer benefit is not good. Talk of a helicopter solved all the problems, and the farmer took action to create the prototype of the large industrial ceiling fan.

In the process of use, the advantages of large industrial ceiling fans began to emerge, its cooling, comfort, ventilation effect, coverage and energy-saving than ordinary small fans are much better。

01

广泛应用于

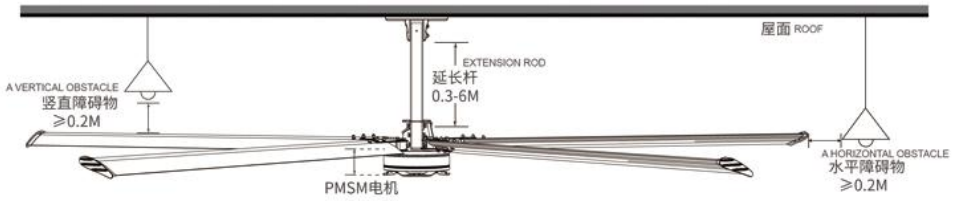
商场、超市/工业厂房/物流仓库
候车室/展览馆/体育馆

Widely used
Shopping malls, supermarkets/industrial buildings/logistics warehouses waiting rooms/exhibition halls/gymnasiums

功能简介

BRIEF INTRODUCTION TO FUNCTIONS

02



人体降温

THE BODY COOLS DOWN

在人员密集公共区域，容易产生闷热感，大型工业吊扇产生的自然微风吹拂在人体上，通过汗液蒸发带走热量，从而使人体降温，带来凉爽的感受，通常，这种降温能达5-8℃.

In crowded public areas, it is easy to feel stuffy. Natural breeze generated by large industrial ceiling fans blows on the human body and carries heat away through evaporation of sweat, thus cooling the human body and bringing a feeling of coolness, this cooling can reach 5-8°C.

干燥除湿

DRY AND DEHUMIDIFY

在车间，仓库等区域，容易产生潮气，大型工业吊扇能促使整个区域空气混合，将湿气、潮气和其它难闻气体进行稀释和降解，从而改善空气质量，防止霉变和细菌滋生。

In the workshop, warehouse and other areas, easy to generate moisture, large industrial ceiling fans can promote the air mixing throughout the region, the moisture, moisture and other unpleasant gases diluted and degraded, thus improving air quality, prevent mildew and bacterial growth.



The humidity in the work area produces mold



The work area is dry and fresh

湿度范围 Humidity range						
正常状况 Normal	30%	40%	50%	60%	70%	80%
打开风扇 Turn on the fan	27%	35%	43%	50%	58%	65%
降低相对湿度 Reduce relative humidity	3%	5%	7%	10%	12%	15%
湿度下降百分比 Percent reduction in humidity	10.00%	12.50%	14.00%	16.67%	17.14%	18.75%

节能降耗

SAVE ENERGY AND REDUCE CONSUMPTION

在商场、超市等场合，如果单靠空调来实现制冷制热的效果，需要的空调机组能耗非常大；但如果空调和大型工业吊扇配合，可把室内冷热空气充分混合均匀，减少空调机组开机时间和开启数目，大幅节约电能。

In shopping malls, supermarkets and other occasions, if only rely on air-conditioning to achieve the effect of refrigeration and heating, the air-conditioning units required very large energy consumption; but if the air-conditioning and large industrial ceiling fans with, the indoor hot and cold air can be fully mixed evenly, reducing the start-up time and the number of air-conditioning unit open, significantly saving electricity



WORKING AREA TEMPERATURE 23°C



WORKING AREA TEMPERATURE 16°C

环流通风

CIRCULATION VENTILATION

在高大宽阔的密闭空间中，由于空气流动性差，一些烟尘对人体不好的物质容易聚集在地面附近，大型工业吊扇能调动室内空气形成环流，从而达到改善室内空气质量的效果。

In the high and wide airtight space, because the air fluidity is bad, some smoke and dust to the human body bad material easy to gather near the ground, the large industrial ceiling fan can mobilize the indoor air to form the circulation, thus, the indoor air quality can be improved.



产品技术优势
TECHNICAL ADVANTAGES OF PRODUCTS

03

1.控制系统技术 CONTROL SYSTEM TECHNOLOGY

控制器硬件功率部分前级增加PFC电路,额定电压输入220VAC，实际使用可兼容110VAC，可适应宽电压输入，改善电压不稳等恶劣条件情况带来的不好的体验效果，而且同样输出功率条件下带PFC电路的输入母线电流可做到更小(比不带PFC电路母线电流小30% 50%左右),因而更加省电；

控制系统软件采用无位置正弦波矢量控制方案，采用电子换向取代传统的电刷机械换向，降低了电机损耗和噪音，增加运行可靠性;同时可以通过检测电机的反电势来计算电机转子位置，无需额外安装霍尔位置传感器,更加简化了电机结构和体积，提高电机使用寿命。

控制柜采用全密封式设计，可有效防尘、防水，提升柜内电子元器件寿命,满足户外使用条件，适应阴雨潮湿等恶劣环境;控制柜操作和调试无需开箱,杜绝了误操作触电风险，更加安全可靠。

控制柜带WIF通讯模块，可远程控制，一方面方便用户操作，同时也可作为后台监控控制系统状态，出现故障能第一时间察觉并快速做出反应;如果客户忘记关机,可远程控制关机。

The PFC circuit is added in the front of the hardware power part of the controller, the rated voltage input is 220VAC, the practical use can be compatible with 110VAC, can adapt to wide voltage input, improve voltage instability and other bad conditions brought about by bad experience effect, at the same output power, the input bus current with PFC circuit can be smaller (about 30% 50% less than the bus current without PFC circuit) , so more power saving; The software of the control system adopts the position-free sinusoidal vector control scheme and electronic commutation instead of the traditional brush mechanical commutation, which reduces the loss and noise of the motor and increases the operation reliability At the same time, the rotor position can be calculated by detecting the back emf of the motor without additional installation of Houle position sensor, which simplifies the structure and volume of the motor, and improves the service life of the motor. The control cabinet adopts full-sealed design, which can effectively prevent dust and water, improve the life of the electronic components in the cabinet, meet the outdoor working conditions, and adapt to the harsh environment such as rain and humidity, it eliminates the risk of electric shock by mistake and is more safe and reliable. Control cabinet with WIF communication module, remote control, on the one hand to facilitate user operation, but also as a background monitoring and control system status, failure can be the first time to detect and respond quickly; If the customer forgets to shut down the machine, the machine can be shut down remotely.

2.自主研发永磁外转子同步电动机 SELF-DEVELOPED PERMANENT MAGNET EXTERNAL ROTOR SYNCHRONOUS MOTOR



✔ 高效节能 HIGH EFFICIENCY AND ENERGY SAVING
✔ 运行可靠 RELIABLE OPERATION
✔ 十年质保 TEN-YEAR WARRANTY

- 迷宫式结构，能有效防止灰尘进入、加速散热
- 升级电机内核散热系统，温升更低、寿命更长
- 永磁同步电动机，高效节能、体积小、重量轻.

- The labyrinth structure can effectively prevent dust from entering and accelerate heat dissipation
- Upgrade the motor core cooling system, lower temperature rise, longer life.
- Brushless electric motor, energy efficient, small size and light weight.



重要特点 IMPORTANT FEATURES

同样都使用变频器，但永磁变频电机不需要无功励磁电流，可以显着提高功率因数，减少了定子电流和定子电阻损耗，而且在稳定运行时没有转子铜耗，进而可以减小风扇(小容 量电机甚至可以去掉风扇)和相应的风摩损耗.

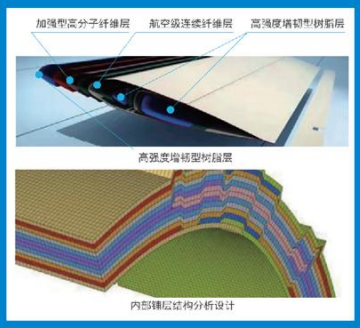
The inverter is also used, but the permanent magnet frequency conversion motor does not need reactive excitation current, can significantly improve the power factor, reduce the stator current and stator resistance loss, and in stable operation, no rotor copper loss, in turn, the fan (small capacity, volume motor can even be removed from the fan) and the corresponding wind friction loss can be reduced.

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3.机翼型航空复合材料风叶 AIRFOIL-TYPE AERO-COMPOSITE BLADES

✔大风量 ✔风力均匀 ✔贴近自然风 ✔噪音小

LARGE WIND, EVEN WIND, CLOSE TO THE NATURAL WIND, LOW NOISE



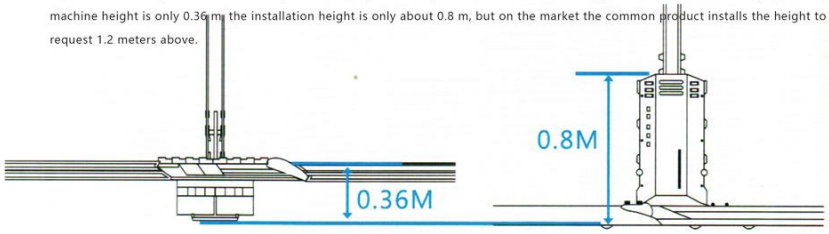
- ●迷宫式结构，能有效防止灰尘进入、加速散热
- ●升级电机内核散热系统，温升更低、寿命更长
- ●永磁同步电动机，高效节能、体积小、重量轻.
- ●安装特殊设计翼尖导流板；
- ●有效减少空气在运动中流线型的扇叶末端形成涡流；
- ●减少涡流造成的能量损失，提高效益；
- ●运转稳定；
- ●有效降低噪声

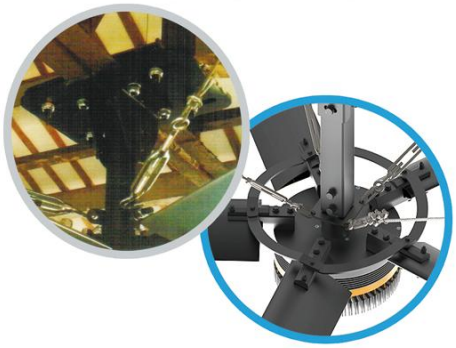
- ●The labyrinth structure can effectively prevent dust from entering and accelerate heat dissipation
- ●Upgrade the motor core cooling system, lower temperature rise, longer life
- ●Brushless electric motor, energy efficient, small size and light weight.
- ●Installation of specially designed wingtip baffles;
- ●Effectively reduce the air in the movement of the end of the streamlined fan to form vortex;
- ●Reduce the energy loss caused by eddy current, improve the efficiency;
- ●Stable operation;
- ●Effective noise reduction

4.安装与可靠性优势 INSTALLATION AND RELIABILITY ADVANTAGES

由于永磁电机本体结构紧凑,需要的安装高度更低,适用场合更广;主机高度仅0.36M，安装高度只需0.8M左右,而市面上常见产品安装高度普遍要求1.2M以上。

Due to the compact structure of the permanent magnet motor, the required installation height is lower and more widely applicable; the main machine height is only 0.36 m, the installation height is only about 0.8 m, but on the market the common product installs the height to generally request 1.2 meters above.





可靠性方面,吊扇主连接轴顶端采用弧形槽连接座,可大幅度前后左右调节延长杆,确保吊扇竖直向下，保证吊扇平衡,避免振动和离心力传递给建筑物；

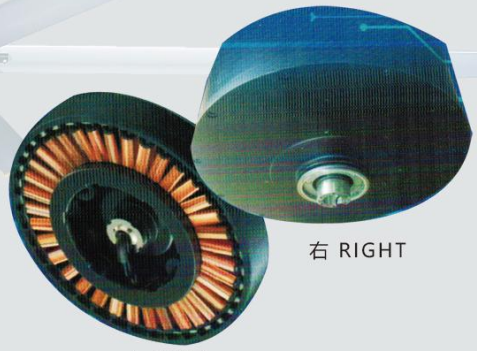
延长杆上连接4条钢索固定在主连接轴四周位置,可以减轻主连接轴位置压力，同时减小吊扇本体的摆动幅度;

For reliability, the top of the main connecting shaft of the ceiling fan adopts the arc-shaped groove connecting seat, which can adjust the lengthening rod to ensure that the ceiling fan is vertical and downward, to ensure the balance of the ceiling fan, to avoid the vibration and centrifugal force transmitted to the building; Four cables connected on the extension bar are fixed around the main connecting shaft, which can reduce the position pressure of the main connecting shaft and the swing amplitudeof the ceiling fan

现有电机中间连接轴往下只延伸到电机下壳盖以内(如图左)；

我司改进的电机中间连接轴伸出电机下壳盖以外(如图右)，可在底盘增加钢索通过中空的连接轴与屋顶固定,增加电机系统的安全系数。

The existing middle connecting shaft of the motor extends down only to the lower cover of the motor (as shown in the left) , and the improved middle connecting shaft of the motor extends beyond the lower cover of the motor (as shown in the right) , steel cables can be added to the chassis and secured to the roof through a hollow connecting shaft to increase the Factor of safety of the motor system.



右 RIGHT

左 LEFT

行业案例

INDUSTRY CASE

05



厂房车间 / 物流仓储 /健身房 / 体育馆 / 室内运动场 / 练功房/会展中心 / 4S店 / 产品集散市场 / 大型商场与超市/企业写字楼的前厅 / 别墅 / 食堂 / 博物馆 / 大型户外商业活动租赁/大型游乐场 / 动物植物园 / 儿童游乐场 /火车站 / 高铁站 / 汽车站 / 地铁站 / 候机楼.....

Factory workshop/logistics storage/gymnasium/gymnasium/Indoor Sports Ground/exercise room/convention and Exhibition Centre/4s shop/product distribution market/lobby/Villa/Canteen/museum/large-scale outdoor commercial activity rental/large playground/zoological and botanical gardens/children's playground/train station/high-speed railway station/bus station/subway station/terminal...

产品系列简介
BRIEF INTRODUCTION OF PRODUCT SERIES

06

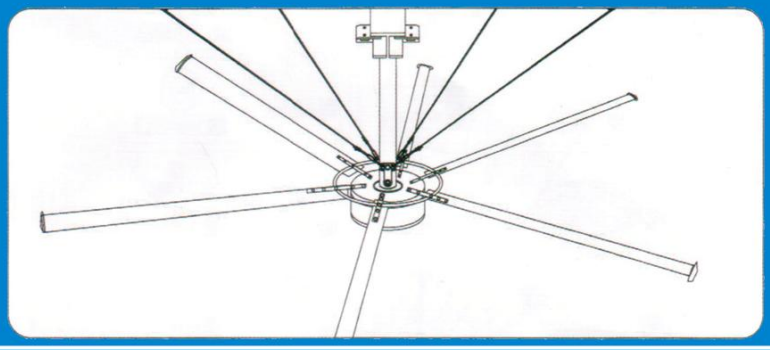
LSD系列 LSD SERIES

LSD系列节能吊扇是由路士达智能科技研发的一种新能源永磁同步电机技术的新型风扇;其最大直径可达7.3米。该产品结合了空气动力学、传动力学、脉宽调制控制技术、机械力学、仿真模拟技术、通讯控制、工业设计等一系列综合学科的先进技术，采用先进智能精密加工设备制造而成。高效节能、气流流动循环，达到人体降温改善环境。

使用场所:工厂、仓库、体育场、机场、车站、健身房、大型超市等。

LSD series energy-saving ceiling fan is a new type of permanent magnet synchronous motor technology developed by lustar intelligent technology; its maximum diameter can reach 7.3 meters. The product combines the advanced technologies of aerodynamics, transmission dynamics, pulse width modulation control technology, mechanical mechanics, simulation technology, communication control, industri al design, etc. , it is made of advanced intelligent precision machining equipment. High-efficiency energy-saving, airflow circulation, to achieve the human body to improve the environment temperature.

Places: factories, warehouses, stadiums, airports, stations, gyms, supermarkets and so on.



产品规格表 PRODUCT SPECIFICATION SHEET

型号 Models	直径（米） Diameter (m)	功率 (KW) Power (KW)	转速 (r/min) Speed (r/min)	输入电压 (v) Input Voltage (V)	最大风量 (m^3/min) Maximum air volume	重量 (kg) Weight (kg)
LSD-1-15A-6Y	7.3	1.5	55	380	14800	110
LSD-1-13A-6Y	6.3	1.3	60	380	13800	100
LSD-1-9A-6Y	5.5	0.9	75	220	13000	80
LSD-1-8A-5Y	5.0	0.8	80	220	11500	70

注:1、重量:重量不包含控制箱，延长杆，顶部连接部件等

2、直径:以上所列为产品标准直径，其他规格需定制。

3、电源:三相380V±10%或单相220V±10%。

4、电机:永磁同步电机

NOTE: 1、Weight: weight does not include control box, extension bar, top connecting parts, etc.

2、Diameter: the above listed as the product standard diameter, other specifications to be customized.

3、Power supply: three-phase 380V ± 10% or single-phase 220V ±10% .

4、Motor: permanent magnet synchronous motor.

产品系列简介
BRIEF INTRODUCTION OF PRODUCT SERIES

07

移动永磁工业风扇
Mobile permanent magnet industrial fan



移动永磁工业风扇直径1M-2.5M，适用广泛场所需求。超远的送风距离，风速有效距离超过30m。4只脚轮的设计，可自由移动至需要的地方，用刹车轮进行固定。节能省电，能耗低至一天只需几块钱。安静低噪，无极调速，更能满足您对风量的精准控制需要。

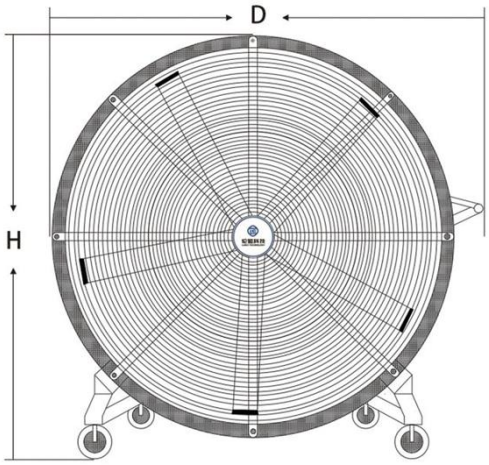
The diameter of the mobile permanent magnet industrial fan is 1M-2.5M,Suitable for a wide range of site needs.Extremely long supply distance,The effective distance of the wind speed is more than 30m.Design with 4 casters,You can move freely where you need it,Fix with brake wheels.Energy saving, energy consumption as low as a few dollars a day.Quiet and low noise,Stepless speed regulation,It can also meet your needs for accurate control of air volume.

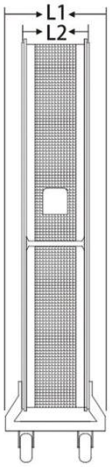
产品参数
Product details

型号 (Models)	电压 (Voltage) V	功率 (Power) W	额定电流 (Current rating) A	直径 (Diameter) M	最大转速 (maximum rotational-speed)PM	风量 (Wind volume) m3/min	重量 (Weight) kg	噪音等级 (noise level) (dB)A
YF-1000-5	220V	500W	2.0A	1.0M	500	580m³/min	45	≤60
YF-1250-5	220V	500W	2.0A	1.25M	450	650m³/min	53	≤60
YF-1500-5	220V	750W	2.0A	1.5M	400	900m³/min	95	≤60
YF-1500-6	220V	750W	2.6A	1.5M	400	1050m³/min	100	≤60
YF-2000-6	220V	950W	2.6A	2.0M	260	1650m³/min	140	≤60

产品系列简介
BRIEF INTRODUCTION OF PRODUCT SERIES

08





产品尺寸
Product dimensions

型号 (Models)	D	H	L1	L2
YF-1000-5	110cm	110cm	51cm	32cm
YF-1250-5	135cm	135cm	51cm	32cm
YF-1500-5	170cm	170cm	51cm	32cm
YF-1500-6	170cm	170cm	51cm	32cm
YF-2000-6	220cm	220cm	51cm	32cm

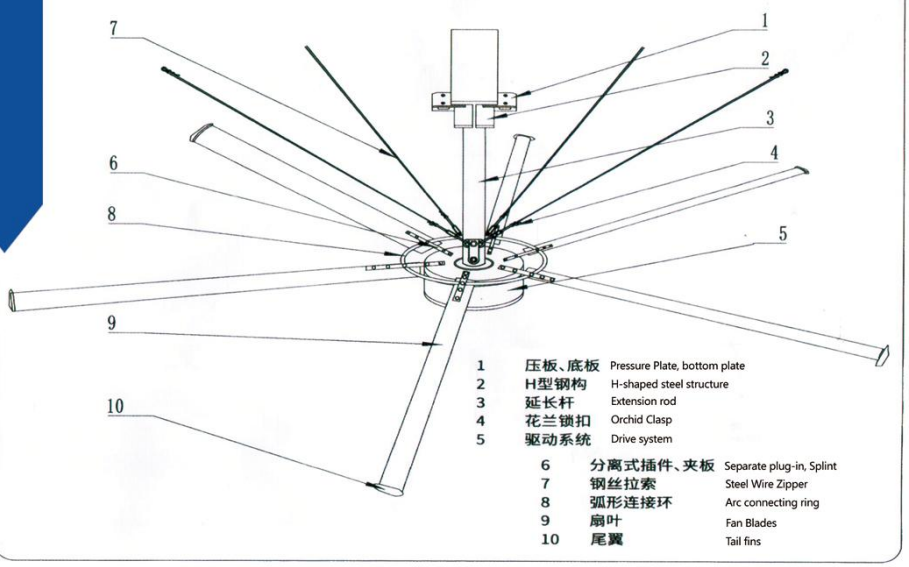
产品系列简介
BRIEF INTRODUCTION OF PRODUCT SERIES



09

移动风扇
Move the fan

产品结构图
PRODUCT STRUCTURE DIAGRAM



1	压板、底板	Pressure Plate, bottom plate
2	H型钢构	H-shaped steel structure
3	延长杆	Extension rod
4	花兰锁扣	Orchid Clasp
5	驱动系统	Drive system
6	分离式插件、夹板	Separate plug-in, Splint
7	钢丝拉索	Steel Wire Zipper
8	弧形连接环	Arc connecting ring
9	扇叶	Fan Blades
10	尾翼	Tail fins

无锡市路士达智能科技有限公司

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Near the Lao Xi Sha Line, Donggang Li Xi Cun, Xishan District, Wuxi City

售后保障

1.质保期内吊扇动力装置出现问题将提供全新的设备，以旧换新。

2.超过保修期，配件以低于市场价提供；

3.我司负责对用户使用，操作维修，保养人员进行免费培训；

1. New equipment will be provided to replace the old ones if problems occur with the ceiling fan power plant during the warranty period.
2. Over warranty, accessories are available at below-market prices.
3. Our Division is responsible for the use of users, operation and maintenance, maintenance personnel for free training;